

Multi-Sensor Calibration Deliverable Document

Document Information

Project: Multi-sensor calibration for a driving simulator

System: Driving simulator with cameras, ToF sensor, and Tobii Pro Fusion eye tracker

Purpose: To document the calibration workflow, coordinate conventions, transformation chain, output files, and validation method used to register all sensors into one global coordinate system.

Final output: A unified extrinsic calibration file containing all sensor-to-global transformation matrices.

1. Project Objective

The objective of this project is to calibrate all sensors mounted on the driving simulator into a common global coordinate system. After calibration, measurements from different sensors can be transformed into the same spatial reference frame and used jointly for later data analysis.

The calibrated sensors include:

- Front camera **Logitech C930e Business Webcam**, referred to as **cam0**
- Right-side camera **Logitech C920 HD Pro Webcam**, referred to as **cam1**
- Rear camera **Logitech BRIO 4K Ultra HD Pro Webcam**, referred to as **cam2 (only Simply for easier measurement)**
- Time-of-Flight camera **ADTF-3175**, referred to as **ToF**
- **Tobii Pro Fusion** eye tracker, using its User Coordinate System, referred to as **Tobii UCS**

The final calibration result provides one rigid transformation matrix for each sensor:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_sensor} \mathbf{P}_{sensor}$$

where $\mathbf{P}_{sensor}$ is a 3D point expressed in the local coordinate system of a sensor, and $\mathbf{P}_{global}$ is the same point expressed in the global coordinate system.

2. Coordinate System Convention

A single global coordinate system was defined and fixed to the rigid body of the driving simulator. This global coordinate system is used as the reference frame for all final sensor poses.

The transformation convention used throughout the project is:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_sensor} \mathbf{P}_{sensor}$$

Each transformation matrix is a 4×4 homogeneous rigid-body transformation:

$$\mathbf{T}_{global_from_sensor} = \begin{bmatrix} \mathbf{R}_{global_from_sensor} & \mathbf{t}_{global_from_sensor} \\ 0 & 1 \end{bmatrix}$$

where:

- $\mathbf{R}_{global_from_sensor}$ is the 3×3 rotation matrix.
- $\mathbf{t}_{global_from_sensor}$ is the 3×1 translation vector.
- The unit of translation is **meter**.
- 3D points are represented in homogeneous form as $[x, y, z, 1]^T$.

The inverse transformation is also stored when needed:

$$\mathbf{P}_{sensor} = \mathbf{T}_{sensor_from_global} \mathbf{P}_{global}$$

with:

$$\mathbf{T}_{sensor_from_global} = (\mathbf{T}_{global_from_sensor})^{-1}$$

3. Calibration Deliverables

The calibration process produces the following deliverables:

3.1 Intrinsic Calibration Files

Each camera has an intrinsic calibration file containing:

- Camera matrix $\mathbf{K}$
- Distortion coefficients
- RMS reprojection error

The camera matrix is defined as:

$$\mathbf{K} = \begin{bmatrix} f_x & c_x \\ & f_y & c_y \\ & & 1 \end{bmatrix}$$

The OpenCV distortion model uses:

$$[k_1, k_2, p_1, p_2, k_3]$$

3.2 Sensor Extrinsic Calibration Files

Individual extrinsic files were generated for intermediate calibration steps, including:

- cam0 to global
- cam1 to cam0
- cam2 to cam0
- ToF to cam0
- Tobii UCS to global
- Composed cam1 to global
- Composed cam2 to global
- Composed ToF to global

3.3 Final Master Extrinsic File

A final master file was generated to store all sensor poses in one place using the same convention:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_sensor} \mathbf{P}_{sensor}$$

The master file contains:

- $\mathbf{T}_{global_from_cam0}$
- $\mathbf{T}_{global_from_cam1}$
- $\mathbf{T}_{global_from_cam2}$
- $\mathbf{T}_{global_from_tof}$
- $\mathbf{T}_{global_from_UCS}$

This file is intended to be used directly in later data analysis.

4. Hardware and Sensor Setup

The sensors were mounted on the driving simulator as follows:

- **cam0:** front-facing camera
- **cam1:** right-side camera
- **cam2:** rear-facing camera
- **ToF:** depth camera mounted near the front camera
- **Tobii Pro Fusion:** eye tracker mounted on the simulator setup

Because each sensor has its own coordinate system, a calibration workflow was required to register all local coordinate frames to the global simulator coordinate frame.

5. Calibration Target

A ChArUco calibration board was used for camera intrinsic and extrinsic calibration. The ChArUco board combines ArUco marker identification with chessboard corner refinement, which improves robustness and corner localization accuracy.

The ChArUco board parameters used in this project were:

Parameter	Value
Number of squares in X direction	5
Number of squares in Y direction	7
Square length	0.03972 m
Marker length	0.02000 m
ArUco dictionary	DICT_6X6_250

The known 3D coordinates of the ChArUco corners were defined in the board coordinate system. These points were matched with detected 2D image corners to estimate board poses relative to each camera.

6. Step 1: Definition of the Global Coordinate System

The global coordinate system was physically defined and fixed to the driving simulator body. This coordinate system served as the reference frame for the full calibration process.

The global frame was defined before estimating any sensor extrinsic parameters. Once defined, all later transformations were composed into this frame.

Choose a point on the driving device as the origin of the global coordinate system. The driver's left-hand direction is the positive x-axis, the direction facing the driver is the positive y-axis, and the vertically upward direction is the positive z-axis. This complies with the right-handed coordinate system, as shown in the figure.

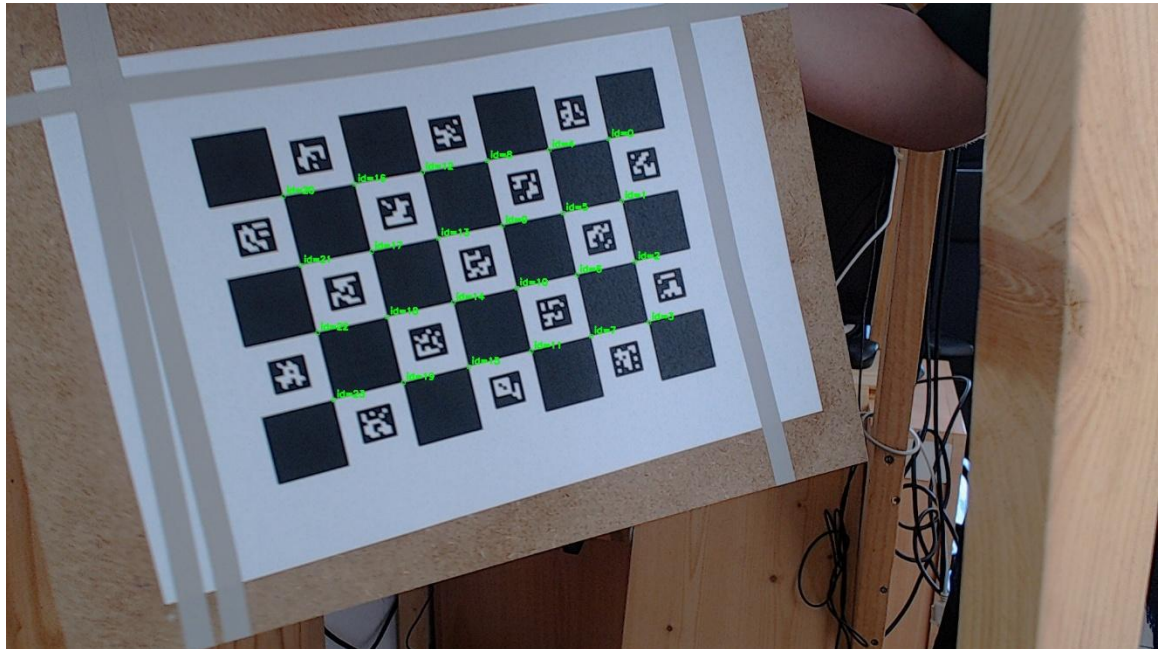


7. Step 2: Intrinsic Calibration of Cameras

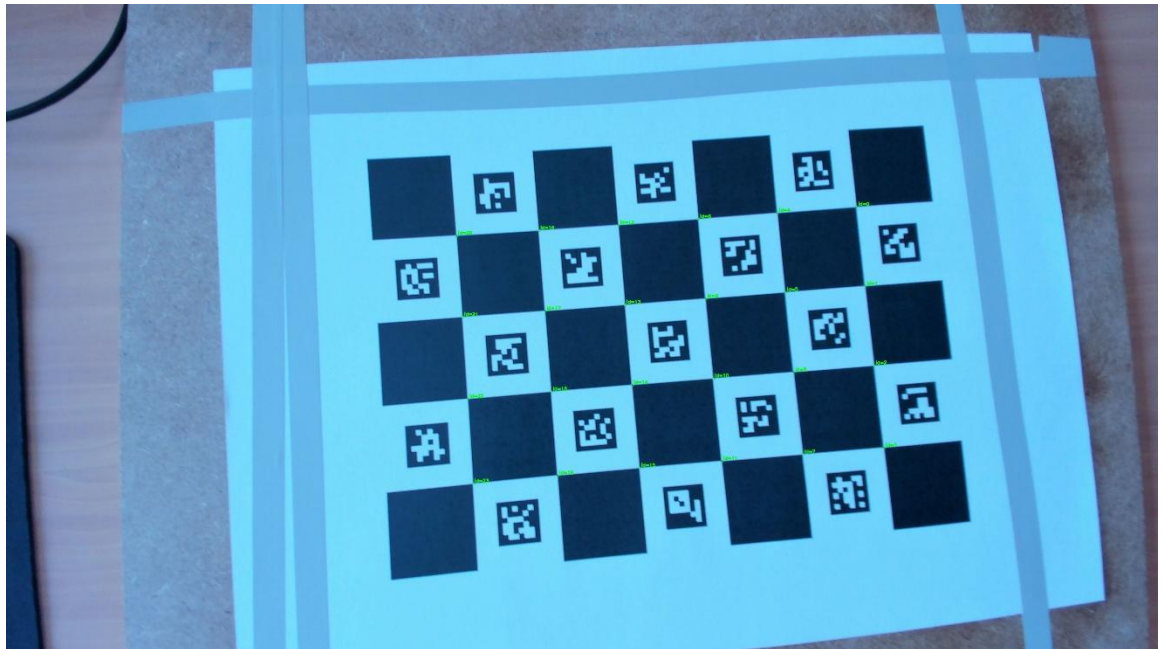
Intrinsic calibration was performed for the following cameras:

- cam0

- cam1



- cam2



- ToF camera

OpenCV calibration tools were used to estimate each camera's intrinsic parameters. For each camera sensor, **30 to 50 images** of the calibration board were captured at various poses to estimate its intrinsic parameters. The output of this step was an intrinsic YAML file for each sensor.

For each camera, the following values were stored:

- Camera matrix

- Distortion coefficients
- RMS reprojection error

The intrinsic calibration files were used as fixed inputs for later extrinsic calibration. During stereo calibration, the intrinsic parameters were fixed to avoid changing previously calibrated camera models.

Notice: Before calibrating the camera's intrinsic parameters, ensure that HDR, autofocus, auto-exposure, and auto-white balance are disabled, and do not modify any other settings. This guarantees that the camera's internal optical structure remains unchanged, thereby ensuring the validity of the calibrated intrinsics.

8. Step 3: cam0 to Global Calibration

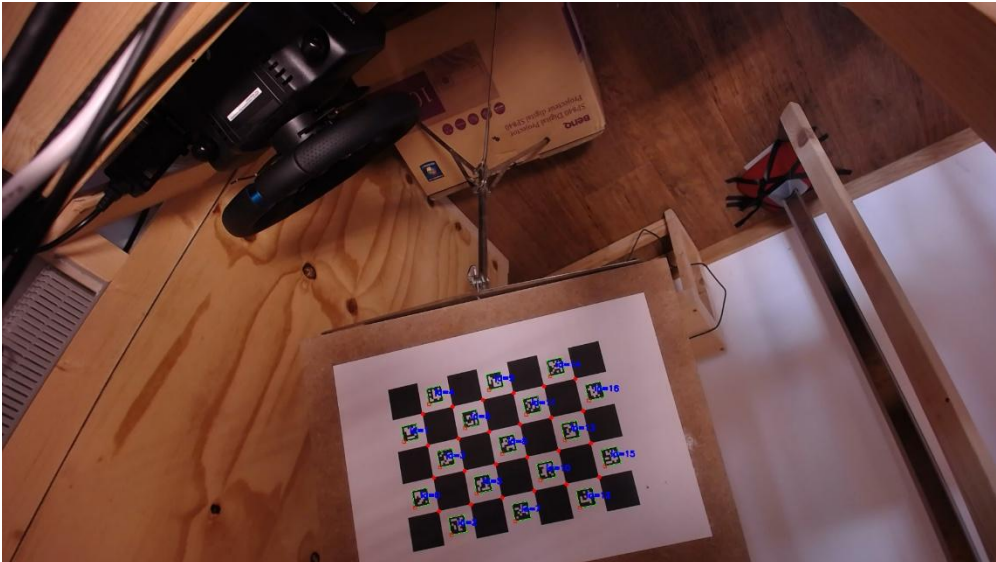
The front camera, cam0, was selected as the first camera to be registered to the global coordinate system.

A ChArUco calibration board was placed in a known pose relative to the global coordinate system. **In physical measurements, a laser distance measure and a tape measure were primarily utilized. By measuring the distance from the laser distance measure to the corners of the calibration board, the inclination angle of the laser in the XY plane (with the laser remaining perpendicular to the X-axis), and the offset distance of the laser's origin along the X-axis, the coordinates of the four calibration board corners in the global coordinate system were computed.**

The ChArUco board was then observed by cam0. From the detected ChArUco corners, the board pose relative to cam0 was estimated.

The cam0-to-global transformation was obtained by:

$$\mathbf{T}_{global_from_cam0} = \mathbf{T}_{global_from_board}(\mathbf{T}_{cam0_from_board})^{-1}$$



Three independent measurements (additional measurements can be performed to further validate the results) were performed. To reduce measurement noise, the final cam0-to-global transformation was computed by averaging the three measurements:

- Translation vectors were averaged directly.
- Rotation matrices were averaged on $SO(3)$, rather than by element-wise matrix averaging.

The result of this step was:

$$\mathbf{T}_{global_from_cam0}$$

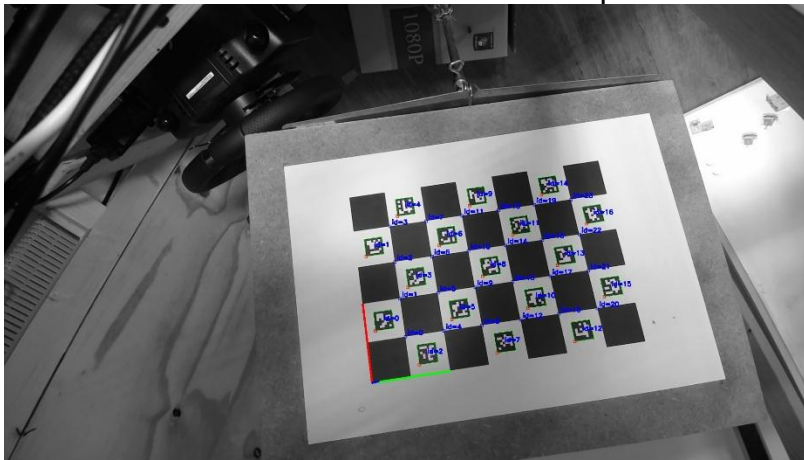
This matrix became the main bridge between the camera calibration chain and the global coordinate system.

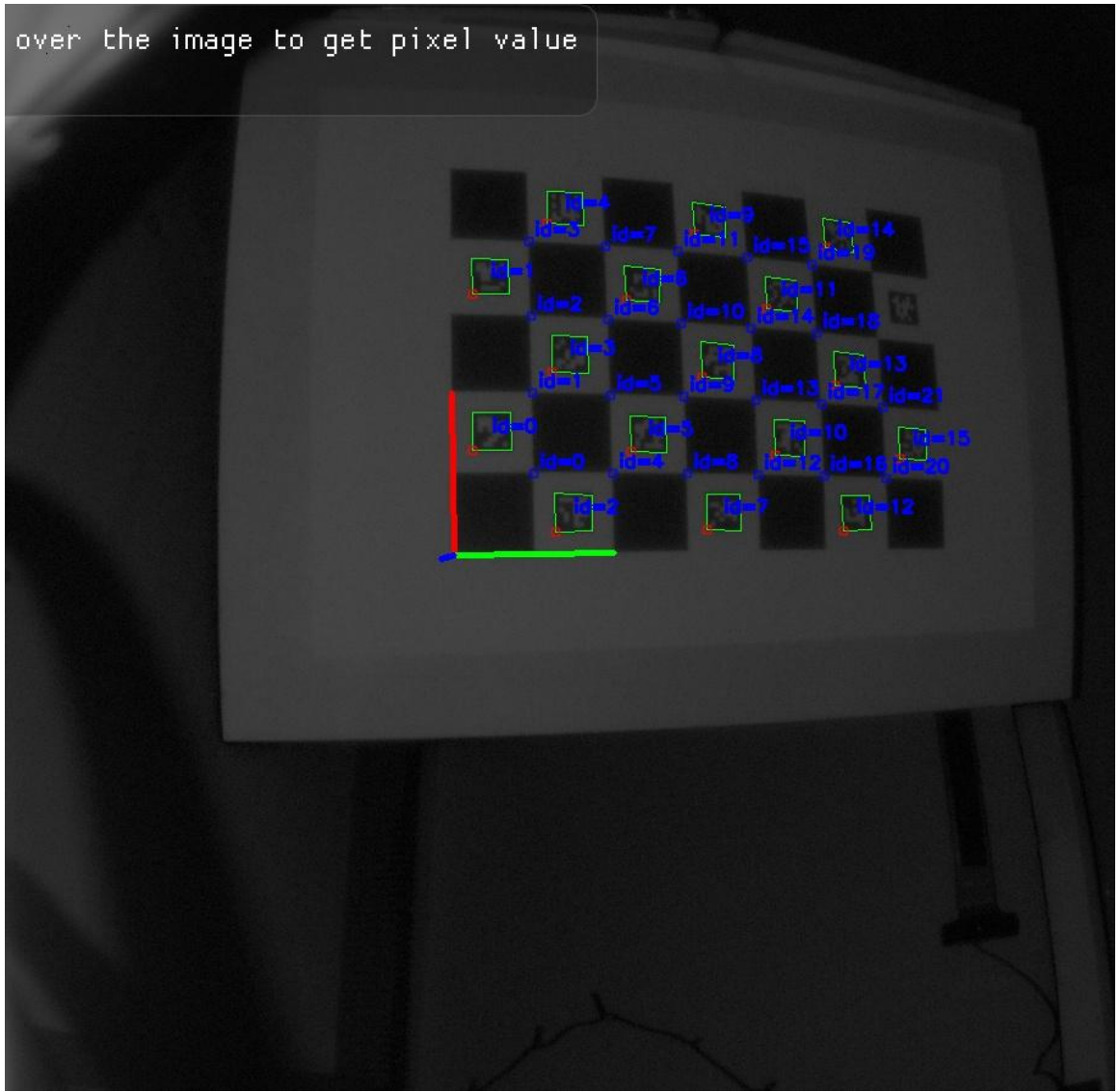
9. Step 4: ToF to Global Calibration

The ToF camera was calibrated relative to cam0 using paired observations of the same ChArUco board (we take **10 pair of photos**, additional measurements can be performed to further validate the results).

For each image pair:

1. The ChArUco board was detected in cam0.
2. The ChArUco board was detected in the ToF image.
3. The board pose was estimated in both coordinate systems.
4. The ToF-to-cam0 transformation was computed.





The relative transformation was calculated as:

$$\mathbf{T}_{cam0_from_tof} = \mathbf{T}_{cam0_from_board}(\mathbf{T}_{tof_from_board})^{-1}$$

After obtaining $\mathbf{T}_{cam0_from_tof}$, the ToF-to-global transformation was computed by matrix composition:

$$\mathbf{T}_{global_from_tof} = \mathbf{T}_{global_from_cam0} \mathbf{T}_{cam0_from_tof}$$

The result of this step was:

$$\mathbf{T}_{global_from_tof}$$

This matrix allows ToF 3D points to be transformed into the global frame.

10. Step 5: cam0, cam1, and cam2 Relative Calibration

The right-side camera and rear camera were registered using camera-to-camera transformations.

The relative calibration between cam0 and cam2 was estimated using stereo ChArUco calibration. The intrinsic parameters of both cameras were fixed, and OpenCV stereo calibration was used to optimize the relative extrinsic transformation. **We used 10 and 14 sets of images for the calibration of cam0-cam2 and cam1-cam2, respectively.**

The stereo calibration outputs a transformation of the form:

$$\mathbf{P}_{cam2} = \mathbf{T}_{cam2_from_cam0} \mathbf{P}_{cam0}$$

The inverse transformation was also computed:

$$\mathbf{P}_{cam0} = \mathbf{T}_{cam0_from_cam2} \mathbf{P}_{cam2}$$

The transformation between cam1 and cam2 was also obtained. Since both cam0 and cam1 were related to cam2, the cam1-to-cam0 transformation was computed through matrix composition:

$$\mathbf{T}_{cam0_from_cam1} = (\mathbf{T}_{cam2_from_cam0})^{-1} \mathbf{T}_{cam2_from_cam1}$$

Then cam1 was transformed into the global coordinate system using:

$$\mathbf{T}_{global_from_cam1} = \mathbf{T}_{global_from_cam0} \mathbf{T}_{cam0_from_cam1}$$

Similarly, cam2 was transformed into the global coordinate system using:

$$\mathbf{T}_{global_from_cam2} = \mathbf{T}_{global_from_cam0} \mathbf{T}_{cam0_from_cam2}$$

The outputs of this step were:

$$\mathbf{T}_{global_from_cam1}$$

and:

$$\mathbf{T}_{global_from_cam2}$$

11. Step 6: Tobii Pro Fusion Eye Tracker Calibration

The Tobii Pro Fusion eye tracker was calibrated using a physical measurement method.

The four corner points on the front surface of the Tobii housing were measured in the global coordinate system. These measured points were used together with the known Tobii User Coordinate System, or UCS, definition to estimate the rigid transformation from the Tobii UCS to the global coordinate system.

The result of this step was:

$$\mathbf{T}_{global_from_UCS}$$

This transformation is used to convert eye position data from the Tobii UCS into the global coordinate system:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_UCS} \mathbf{P}_{UCS}$$

In this project, the main interest was the 3D eye position rather than only the screen gaze point. Therefore, the Tobii calibration focused on locating the eye tracker UCS in the global coordinate system.

12. Step 7: Reprojection Error Validation

Reprojection error analysis was used to validate camera calibration quality.

For a detected ChArUco corner, the reprojection error is defined as:

$$e_i = \sqrt{(u_i - \hat{u}_i)^2 + (v_i - \hat{v}_i)^2}$$

where:

- (u_i, v_i) is the detected image point.
- $(\hat{u}_i, \hat{v}_i)$ is the projected image point.
- e_i is the pixel reprojection error.

Two types of reprojection errors were evaluated:

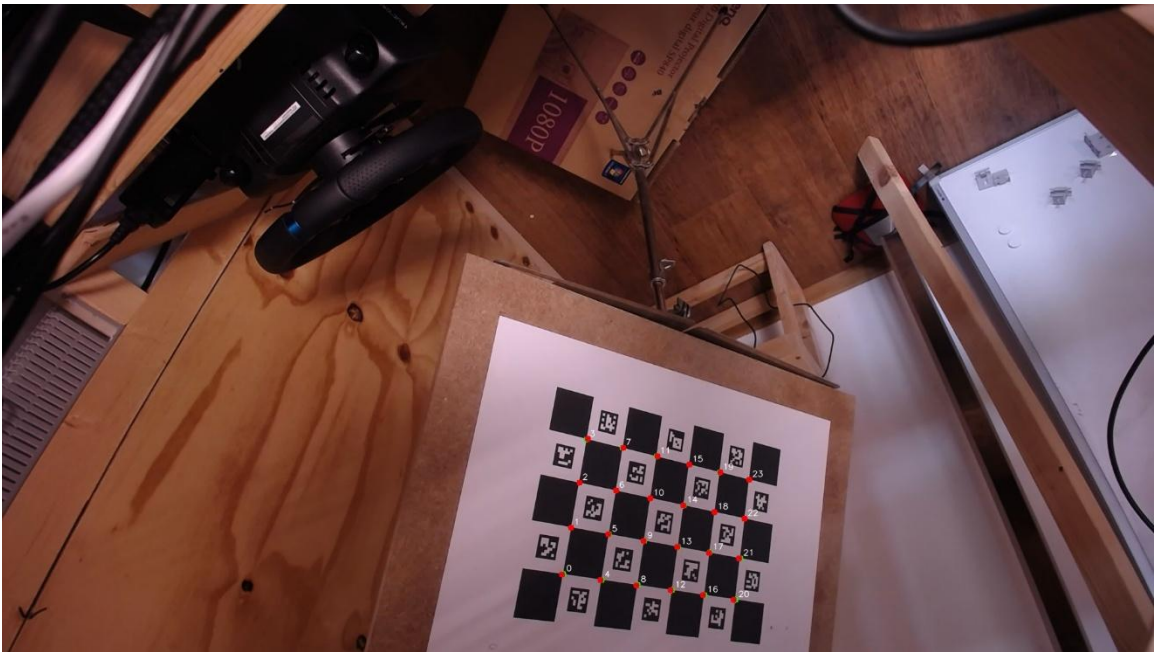
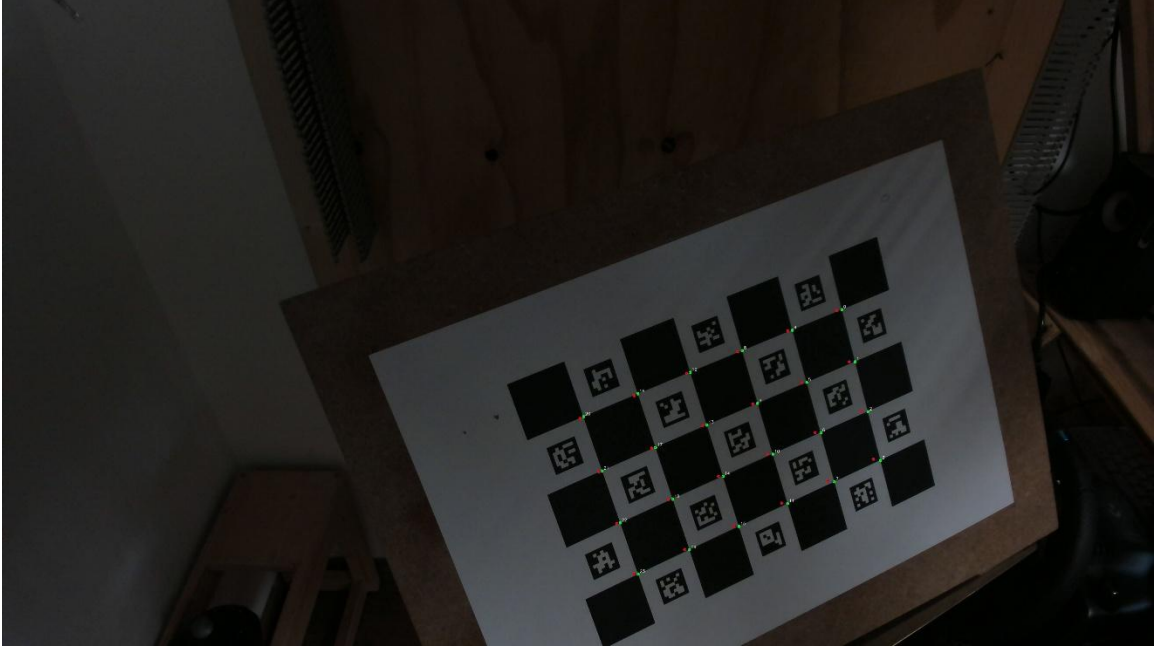
12.1 Self-Reprojection Error

Self-reprojection error was used to evaluate whether each camera's intrinsic calibration and single-camera pose estimation were reasonable.

12.2 Cross-Camera Reprojection Error

Cross-camera reprojection error was used to evaluate whether the relative extrinsic transformation between two cameras was accurate.

For camera-to-camera calibration, stereo calibration RMS and cross-camera reprojection errors were used as quality indicators.



13. Final Master Extrinsic File

After all individual and composed transformations were obtained, they were merged into a single master extrinsic file.

The master file uses the same convention for every sensor:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_sensor} \mathbf{P}_{sensor}$$

The file stores:

- Sensor name
- Source file
- Source matrix node
- Transformation from sensor to global
- Inverse transformation from global to sensor
- Rotation matrix
- Translation vector in meters and centimeters
- Basic rotation matrix quality checks, including determinant and orthogonality error

This file is the main calibration deliverable for downstream data analysis.

14. Data Analysis Usage

During later data analysis, all 3D sensor measurements should be converted to homogeneous coordinates before applying the corresponding transformation.

For a point measured by a sensor:

$$\mathbf{P}_{sensor} = [x, y, z, 1]^T$$

The point is transformed into the global coordinate system by:

$$\mathbf{P}_{global} = \mathbf{T}_{global_from_sensor} \mathbf{P}_{sensor}$$

The first three elements of $\mathbf{P}_{global}$ are the 3D coordinates of the point in the global frame.

15. Quality Checks

The following checks were performed or are recommended before using the calibration results:

1. **Rotation matrix validity**
The determinant of each rotation matrix should be close to 1.
2. **Orthogonality check**
 $\mathbf{R}^T \mathbf{R}$ should be close to the identity matrix.
3. **Physical plausibility check**
Translation vectors should be consistent with the physical sensor positions on the simulator.

4. **Reprojection error check**

Camera-to-camera calibration should be evaluated using reprojection error.

16. Limitations

Possible limitations include:

- Physical measurement errors when defining the global coordinate system or Tobii UCS.
- Residual reprojection error in camera-to-camera calibration.
- Mechanical movement of sensors after calibration.
- Reduced accuracy if the calibration board poses are not sufficiently diverse.